Vibe Shuffle: Baseline-Relative Multimodal Sensing for Adaptive Next-Song Selection

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Abstract

Music recommender systems learn durable preferences, but the track that fits can change from one moment to the next. *Vibe Shuffle* examines whether lightweight camera and cardiac observations can inform next-song selection without being presented as a readout of emotion. It expresses each channel as change from a personal reference, collects successive relative Valence–Arousal points during playback, and uses their mean to select the nearest eligible track from the matching region of a fixed catalog. An exploratory within-participant crossover compared this policy with Random selection from the same catalog. Mood fit was higher on average under Vibe Shuffle, while liking changed little; both estimates were uncertain and did not establish a recommendation benefit. The project provides an inspectable research system that connects personal references, multimodal observations, and song choice. It also defines the validation needed before evaluation as a session-aware re-ranking layer in a music service.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Information systems** → *Recommender systems.*

Keywords

affective computing, music recommendation, relative affect, valence and arousal, heart-rate variability, facial cues, baseline-relative adaptation

1 Motivation

1.1 Momentary fit in music recommendation

Music selection presents a temporal mismatch. Recommender systems learn what a listener usually prefers, whereas the track that fits may depend on the present activity, recent listening, and level of activation. People already use music to regulate mood and arousal [11, 25], and contextual music-recommendation research recognizes that stable preference signals cannot describe every listening situation [12, 26]. An energetic track may support exercise and disturb a quiet break. The problem is the gap between general taste and what fits now.

Explicit mood selection can interrupt an otherwise continuous experience. Automatic adaptation offers another route, but its inputs are uncertain. Facial movement does not uniquely reveal felt emotion, and cardiac timing also changes with respiration, posture, activity, fitness, medication, and sensor quality [5, 23, 27]. Vibe Shuffle studies a narrower question: whether changes from a listener's own recent references can guide the next song through a fixed, inspectable rule. The camera is a visual-behavioral channel and the wearable is a physiological channel. Neither assigns an emotion label.

1.2 Conceptual foundation

Dimensional affect models provide a shared vocabulary for listeners and music. Russell's circumplex describes Valence, from unpleasant to pleasant, and Arousal, from low to high activation [22, 24]. Thayer's account adds complementary categories for energetic and tense activation [28]. Both dimensional and categorical models occur in music-emotion research, and their interpretation depends on the chosen labels and stimuli [7]. Vibe Shuffle adopts this two-axis vocabulary as an interface coordinate, not as a complete model of experience.

Figure 1 relates the continuous coordinate, category vocabulary, and HRV-frequency context. Its emotion words and coherence profiles are conceptual; the selection policy does not output them. Each listener point records change from a personal reference. This choice addresses variation across people and contexts without making the signals emotion-specific. Facial judgments depend on body context, culture, and individual expression [2, 14]. Personal centering changes the comparison, not the meaning of the measurement.

Songs occupy a parallel Valence–Energy space. Valence describes conveyed positivity, and Energy describes intensity and activity. Prior interfaces have used these features as catalog-side counterparts of affective Valence and Arousal [9]. Vibe Shuffle follows that convention while keeping the constructs distinct: musical Energy is not physiological Arousal. Music-emotion coordinates depend on subjective annotation and compress a course that changes over time [1, 30]. The system uses them as a controlled retrieval space rather than ground truth.

1.3 Related work

Prior music systems connect listener information with song coordinates in several ways. *Feel the Moosic* lets the listener choose a position in a simplified affect space and requests tracks from corresponding Spotify Valence–Energy ranges [9]. Janssen et al. learned listener-specific relations between music and physiological response and used them to steer an affective music player

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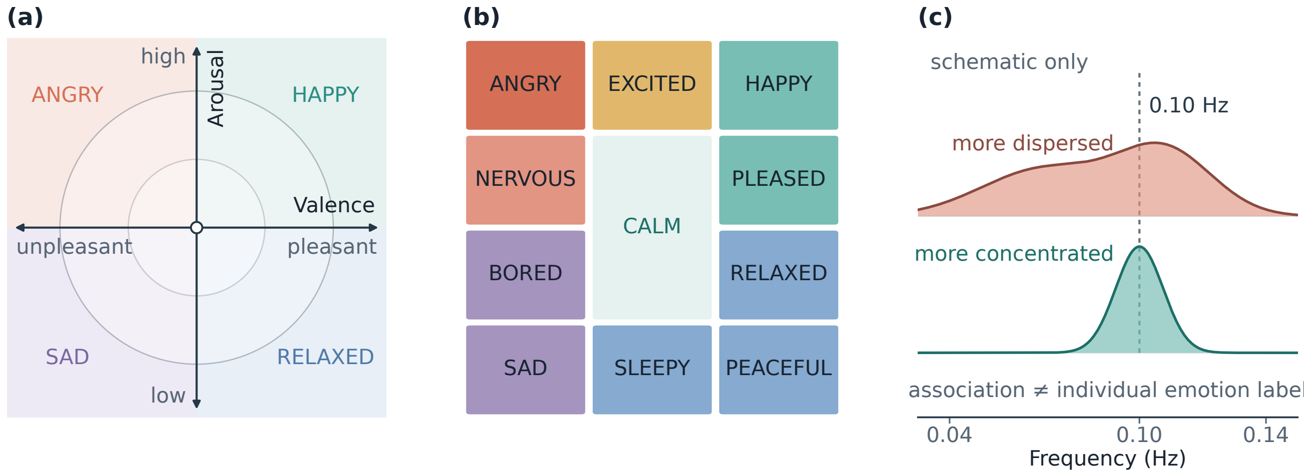

Figure 1: Conceptual foundations. Panels (a)–(b) are original redrawings adapted from Helmholz et al. [9], who operationalized Russell's circumplex [24] and Thayer's mood–arousal model [28]. Panel (c) is an original schematic adapted from the group-level frequency distinction reported by Balaji et al. [4]; it contains no empirical observations.

toward a goal [10]. Ayata et al. estimated Valence and Arousal from wearable galvanic-skin-response and photoplethysmography measurements [3], while Tran et al. matched a facial estimate with nearby tracks in Spotify feature space [29]. These systems establish several routes from explicit or inferred context to music.

Two issues remain. First, classification accuracy does not establish that a recommended track improves the listening experience. Second, the path from sensor observation to selection can hide consequential assumptions. Explanations may support transparency, trust, effectiveness, or decision-making [21], while user experience also depends on effort, privacy, and context [13]. A reviewable policy should expose what was observed, how it was summarized, and why one song became eligible.

1.4 Research gap, questions, and contributions

Among the systems reviewed here, few expose a baseline-relative path from multimodal observations to a reviewable next-song decision. It also remains unclear whether that policy changes listener judgments when catalog, exposure, and rating procedure are held constant. Vibe Shuffle addresses this gap with a personal reference, an event-driven listening trajectory, a fixed region-and-distance rule, and a Random comparator drawn from the same frozen catalog.

RQ1. How can within-listener changes in facial and cardiac observations be transformed into an inspectable next-song decision without assigning absolute emotion labels?

RQ2. How does Vibe Shuffle compare with Random song selection in participants' ratings of mood fit and liking?

The paper makes three concrete contributions. First, it defines a baseline-relative formulation of multimodal music adaptation. Second, it specifies the path from personal references and listening trajectories to an eligible-song decision. Third, it reports a matched exploratory comparison that quantifies the direction and uncertainty of mood-fit and liking differences. These contributions define a testable policy and its current evidence boundary.

In the exploratory comparison, mood-fit ratings were 0.56 points higher under Vibe Shuffle on average, but the confidence interval crossed zero. Liking differed by 0.09 points and was similarly uncertain. These estimates motivate a better instrumented test of the policy; they do not show that the system improves listening.

2 Concept Overview

2.1 Design objectives

Three objectives guided the concept. First, interpretation should be personal: observations are expressed as changes from the listener's own reference rather than population thresholds. Second, the decision should remain inspectable: the research record should expose the path from observations to the selected song. Third, adaptation should be conservative: sensor outputs guide recommendation but are not shown as the listener's emotion.

The project tests matching rather than mood regulation. It does not assume that an energetic track should calm an activated listener or that a positive track should repair negative Valence. A later system could offer maintaining, intensifying, or shifting as explicit goals. Combining them in the present policy would make a successful recommendation difficult to interpret.

2.2 Ideation and refinement

The repository begins with an integrated research system and does not preserve sketches, interviews, or a dated sequence of early prototypes. This section reconstructs only the design decisions visible in the final artifact. A personal reference bounds interpretation to within-listener change. Successive observations form a trajectory instead of reducing a track to one frame or cardiac sample. A fixed region-and-distance rule makes the resulting choice inspectable.

Finally, a Random condition from the same catalog separates the adaptive rule from exposure count and rating procedure.

A later artifact and manuscript audit clarified the claim boundary, separated diagnostic coherence from the selection rule, documented result provenance, and exposed unresolved implementation and study limitations. These are specification and reporting refinements, not participant-driven redesigns; the archive contains no qualitative feedback log.

The research system also preserves graceful degradation. Camera and cardiac inputs are optional, the export records which evidence path was available, and adaptation waits until the next track. These choices keep a session operable when signals are missing while making missingness part of the analysis rather than hiding it behind a recommendation score.

2.3 Final concept and intended use

Vibe Shuffle follows a six-step loop: acquire personal references, collect camera and cardiac observations during playback, express them as relative Valence–Arousal points, summarize the trajectory, choose the nearest eligible song in the corresponding catalog region, and record the rating and decision trace. Adaptation occurs between tracks, so brief fluctuations cannot interrupt the current song. Figure 2 shows this loop.

The intended use is scientific evaluation of baseline-relative adaptation. A future music service could test the policy as a reranking layer above a conventional preference model. The present project neither implements nor validates that deployment.

3 Signal & Pipeline

3.1 Personal references and measurement boundary

The multimodal input has two distinct channels. A camera supplies visual and behavioral cues; an optional wearable supplies physiological timing. The camera reference runs for up to 14 seconds and may be skipped. The cardiac reference records 120 seconds of device-produced heart rate and RR intervals. Subsequent observations are centered on these personal values. The references do not define a neutral or true emotional state; they provide local comparison points for the session.

The channels were chosen because they are accessible through commodity client-side interfaces and offer complementary evidence: visible behavior is used primarily for relative Valence, while cardiac change is used primarily for relative activation. Their accessibility makes them practical for an HCI research system, without making them inherently valid measures of emotion. Evaluation must test the complete decision and validate each mapping separately.

3.2 Camera-derived relative Valence

MediaPipe Face Landmarker runs locally and returns facial blendshape activations [17]. A fixed heuristic combines selected smile, cheek, brow, eye, jaw, frown, and mouth cues into positive, negative, and tension-related scores. Samples scheduled at 120 ms intervals are debiased against the reference and smoothed across frames. More positive visible evidence moves relative Valence right; negative or tension-related evidence moves it left. A slowly adapting reference follows longer changes. If no face is detected, Valence returns to the center.

Visible movement and nodding can also add positive-Valence and upward-Arousal evidence. The channel represents change in observable behavior under a fixed rule, not felt emotion. Lighting, pose, occlusion, morphology, culture, deliberate expression, dancing, and repositioning may change the output [2, 5, 14]. The complete coefficients and gates are retained in `src/expressionModel.js`.

3.3 Cardiac relative Arousal

A compatible Bluetooth Heart Rate Service device supplies beats per minute and, when available, RR intervals. The research system has no raw electrocardiogram and cannot inspect the device's beat detector. Implausible or abruptly changing intervals are rejected before variability is calculated. The reference stores median heart rate, the root mean square of successive differences (RMSSD), and robust spread estimates.

During listening, higher-than-reference heart rate and lower-than-reference RMSSD provide upward Arousal evidence; changes in the opposite direction provide downward evidence. Heart rate receives the larger influence, and RMSSD is damped when the indicators disagree. At least 20 accepted RR intervals are required for the full estimate. RMSSD is a recognized short-term variability summary, but duration, respiration, posture, movement, fitness, medication, contact quality, and vendor processing affect its interpretation [15, 23, 27]. Agreement between 120-second and longer RMSSD measurements has been reported under controlled resting conditions, not short listening windows with movement [19].

The export also records SDNN and a custom spectral-peakedness field called physiology coherence. It differs from the coherence measure associated with post-session emotion labels in large-scale biofeedback data [4]. It is diagnostic only and does not affect song selection.

3.4 Fusion, quality, and local data flow

The pipeline is acquisition, plausibility filtering, personal-reference centering, smoothing, quality assessment, and coordinate fusion. Usable cardiac evidence sets the Arousal base, while camera movement can add upward evidence. Camera movement supplies an upward-only fallback when cardiac data are unavailable. If neither channel is usable, the coordinate returns to the central reference point.

Raw camera frames, facial landmarks, and Bluetooth notifications remain in browser memory. A locally downloaded CSV contains derived summaries, reference status, signal source, quality flags, timing, selected-song metadata, and ratings. It excludes frames, landmarks, raw packets, waveforms, complete trajectories, personal facial-reference values, and the session seed. This is a data-minimizing local export, not a formally privacy-preserving system. Spotify playback, MediaPipe assets, and hosting still require network traffic.

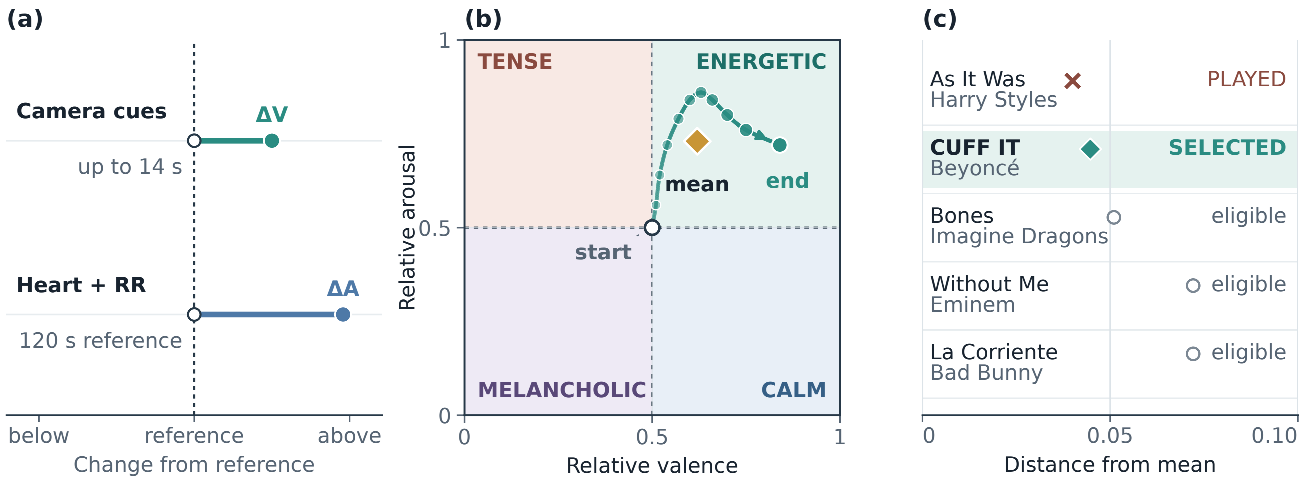

Figure 2: Baseline-relative sensing, trajectory aggregation, and catalog selection. (a) Signed camera and cardiac changes use personal references. (b) The arithmetic mean (gold diamond) summarizes the trajectory between its open start and filled endpoint. (c) *As It Was* is nearest but hypothetically marked as played, so *CUFF IT* is the nearest eligible choice. Panels (a)–(b) are schematic; panel (c) uses catalog-derived candidates and distances with an illustrative played state. Coherence is logged but not used for selection.

4 Inference & Adaptation

4.1 Listening trajectory

During confirmed playback, a change in the fused coordinate appends a point to the trial trajectory

$$\mathcal{T} = \{(V_1, A_1), \ldots, (V_m, A_m)\}.$$

At rating time, the arithmetic mean

$$(\bar{V}, \bar{A}) = \frac{1}{m}\sum_{i=1}^{m}(V_i, A_i)$$

becomes the target for the next Vibe choice. Aggregation reduces dependence on one observation but is event-weighted, because points are added when the coordinate changes rather than at a fixed rate. The export records sample count and listening duration for review.

4.2 Recommendation regions and catalog

Splitting both axes at 0.5 produces four catalog regions: Energetic, Calm, Tense, and Melancholic. These names identify Valence–Energy partitions; they do not classify the participant. Listener Arousal and song Energy are operational counterparts for retrieval, not equivalent constructs [9, 30].

The frozen catalog contains 100 unique Spotify tracks, with 25 in each region. Its retained Valence and Energy coordinates originate from a public historical dataset [18]. The repository preserves the track records but not the download date or a script reproducing their curation. The preserved catalog is a frozen stimulus set, not a fresh derivation.

4.3 Selection policy and Random comparator

Both conditions exclude the current and all previously played tracks. Vibe keeps unplayed tracks in the target region and selects the smallest Euclidean distance:

$$s^* = \arg\min_{s \in Q(\bar{V}, \bar{A})} \sqrt{(V_s - \bar{V})^2 + (E_s - \bar{A})^2}.$$

A session seed resolves exact ties. If the region is exhausted, the policy searches the complete unplayed pool. Random uses the same pool and exclusions but ranks candidates by a session-seeded pseudorandom score. Catalog, playback path, exposure count, and rating questions remain constant, isolating the selection rule more closely than a comparison between different interfaces.

The first track is selected from the setup-time estimate before the facial reference finishes. Later choices use the preceding listening window. Because sensing is optional, a Vibe decision can be camera-only, cardiac-only, fused, or no-signal. Historical participant summaries do not report how often each path occurred.

4.4 Decision trace and prospective service role

Adaptation changes only the next-song ranking. The completed trajectory produces a target, region, eligible set, distance ranking, and selected track. The local record exposes the final coordinate and decision, although it cannot replay every sensing step.

In a future service, a long-term preference model could first generate a suitable candidate pool. A confidence-gated Vibe policy could then adjust the ordering for the current session while preserving an override and a visible explanation. This is a proposed deployment model, not an evaluated feature of the present system.

This design answers RQ1 operationally. Within-listener observations become an inspectable decision because every transformation has a named output: personal reference, relative coordinate, trajectory mean, catalog region, eligible set, distance, and chosen track. A service implementation should expose the reason for a re-ranking, suppress adaptation when signal quality is insufficient, and let the listener return immediately to preference-only selection.

5 User Study & Reflection

5.1 Study design and evaluation plan

The exploratory study used a within-participant crossover with one five-track Random block and one five-track Vibe block. Both conditions drew from the same frozen catalog. Two protocols reversed block order, and the current interface uses odd/even participant numbers to suggest an assignment. Participant-facing mode replaces condition names with protocol identifiers, although masking depends on study administration and is not enforced by the interface. The study was not double blind. Reversing order can reduce, but not remove, practice, fatigue, and carryover risks in a paired design [8].

Each track plays for up to 60 seconds, with an option to rate earlier. After each track, participants answer *How much do you like this song?* and *How well did it fit your current mood?* on seven-point scales. They also select Energetic, Calm, Tense, Melancholic, or Neutral as a categorical mood report. Mood fit measures the proposed contextual match; liking distinguishes that judgment from general song preference. Listening duration, early rating, condition order, and the current implementation's selection fields are exported.

The current protocol specifies the following sequence. After Spotify authentication, the experimenter selects a masked order protocol and pseudonymous participant number. The participant completes the available camera and cardiac references, hears five tracks in the first condition, rates each track, and then completes five tracks under the second condition. Vibe uses the preceding listening window for later adaptive choices; Random ignores the trajectory. The local study file is downloaded when the protocol finishes and can also be saved manually. Because the exact historical software revision is unavailable, this sequence describes the present protocol rather than independently proving every pilot-session step.

5.2 Available evidence and analysis

The retained analysis data contain 15 participant/session rows under generic identifiers. Each row reports one five-trial mean per condition for mood fit and liking. Trial rows, participant characteristics, recruitment and compensation records, consent and ethics documentation, study dates, hardware, room conditions, administered condition order, and a code revision tied to collection are unavailable. These gaps prevent reconstruction of trial completeness, exclusions, within-session variation, or the frequency of camera-only, cardiac-only, fused, and no-signal decisions.

Paired differences are Vibe minus Random, and the participant/session five-trial mean is the unit of analysis. We report condition means and standard deviations, paired mean differences, two-sided 95% confidence intervals, standardized paired effects, and the supplied directional paired tests. Confidence intervals use the paired t distribution with 14 degrees of freedom. Figure 3 adds 20,000 fixed-seed bootstrap resamples to show the mean distribution; its interval bars remain t-based.

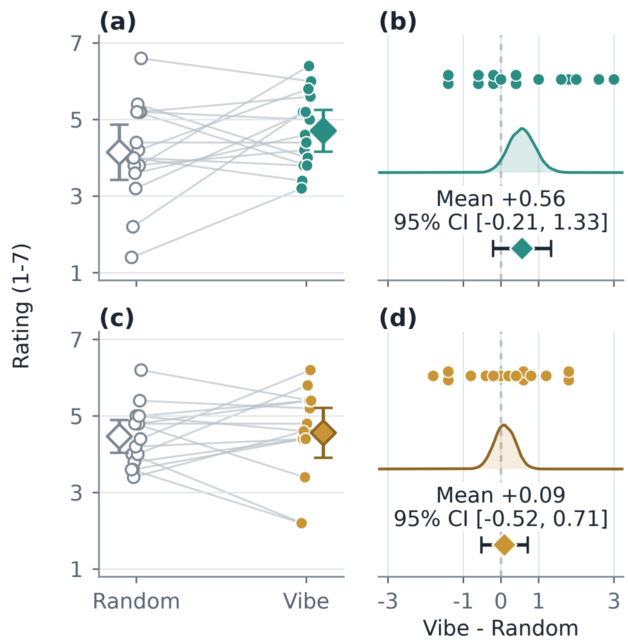

Figure 3: Paired participant/session outcomes. Panels (a)–(b) show mood fit; panels (c)–(d) show liking. Left panels connect Random and Vibe five-trial means and summarize condition means. Right panels show all paired differences, fixed-seed bootstrap mean distributions, and reported paired means with t-based 95% CIs. Both paired intervals cross zero.

The archived means are rounded to one decimal place and reproduce the supplied condition summaries and paired t results to rounding. Exact signed-rank values depend on unavailable precision and tie handling, so we retain the supplied values. No dated preregistration or confirmatory threshold was available. Estimation and two-sided uncertainty are primary; the one-sided tests are reported descriptively [6, 20].

5.3 Results

Mood fit averaged 4.71 under Vibe and 4.15 under Random (Table 1). The paired difference was +0.56 rating points, with a 95% confidence interval from −0.21 to 1.33 and a standardized paired effect of $d_z = 0.40$. Eight participant/session means were higher under Vibe, one was tied, and six were lower. The retained directional paired test yielded $t(14) = 1.56$, $p = .070$; the supplied signed-rank sensitivity check yielded $W^+ = 74.0$, $p = .088$. The interval includes effects on both sides of zero.

Liking averaged 4.56 under Vibe and 4.47 under Random. The paired difference was +0.09, with a 95% confidence interval from −0.52 to 0.71 and $d_z = 0.08$. The directional tests yielded $t(14) = 0.32$, $p = .375$, and $W^+ = 58.5$, $p = .353$. No equivalence margin was specified, so the near-zero estimate does not establish equivalence.

The pilot does not answer RQ2 decisively. Mood-fit ratings were higher on average under Vibe Shuffle, whereas liking changed little, but neither estimate excludes zero or establishes a recommendation benefit.

Table 1: Exploratory summaries of 15 participant/session five-trial means. Confidence intervals are two-sided; retained tests use one-sided Vibe-over-Random alternatives.

Outcome	Vibe M (SD)	Random M (SD)	Difference [95% CI]	$t(14)$, p	W^+, p	d_z
Mood fit	4.71 (0.99)	4.15 (1.30)	+0.56 [−0.21, 1.33]	1.56, .070	74.0, .088	0.40
Liking	4.56 (1.18)	4.47 (0.77)	+0.09 [−0.52, 0.71]	0.32, .375	58.5, .353	0.08

The paired display also shows heterogeneity that a condition mean would hide. Several participant/session differences exceed one rating point in either direction. That variation could reflect music familiarity, starting context, order, signal availability, or genuine differences in how useful matching is. The retained means cannot separate these explanations, but they argue against treating one average response as a universal personalization effect.

5.4 Reflection and design refinement

The study and subsequent artifact audit clarify what the project can support. Personal references provide a local comparison point, but they do not validate the facial or cardiac mappings. A trajectory makes the decision easier to inspect, but its arithmetic mean gives more weight to periods with more coordinate changes. The matched Random condition controls catalog size, playback path, exposure count, and rating procedure; selected songs can still differ in artist, genre, language, familiarity, and popularity.

The audit produced concrete refinements to the research specification. Emotion-detection language was replaced with baseline-relative evidence, diagnostic coherence was separated from the selection policy, and the decision record was defined around target, region, distance, signal source, and quality flags. These changes improve interpretability without strengthening the historical outcome evidence. No qualitative feedback archive supports claims of participant-driven redesign.

The signal-processing and selection rules are deterministic JavaScript modules. Sixty-three tests cover catalog size and balance, export finiteness and quality flags, facial-rule behavior, RR filtering, baseline-relative cardiac behavior, and Random/Vibe selection. They validate the present modules, not sensor accuracy, Spotify playback, the end-to-end study flow, or the code run during the pilot.

5.5 Limitations and threats to validity

Three limitations define the evidence boundary. First, the facial and cardiac mappings are fixed heuristics rather than independently validated measures. Movement may alter camera evidence, raise heart rate, and degrade RR quality at the same time, allowing one behavior to influence both axes. Short HRV windows and unmeasured respiration further constrain interpretation.

Second, the small exploratory crossover and incomplete archive prevent trial-level, order, carryover, missing-data, and subgroup analyses. The retained files contain no recruitment, ethics, hardware, or trial-level records, so those parts of the study cannot be reconstructed. One-sided tests should not be treated as confirmatory without preregistration. The first adaptive choice may precede completed references, sensing is optional, and the historical summaries do not identify the signal path behind each choice.

Third, the frozen catalog improves experimental control but does not represent a dynamic streaming service. The conditions share a catalog, not identical tracks. The system also uses event-weighted rather than time-weighted aggregation. At the exact no-signal center, current code paths disagree over whether the boundary belongs to Calm or Energetic. That edge case should be unified before another study.

5.6 Next evaluation

The next evaluation should begin with controlled changes in expression, lighting, posture, movement, and respiration applied to the frozen mappings. An instrumented pilot should then require completed references, timing and missing-data flags, and a complete ten-trial record. Camera-only, cardiac-only, fused, and Random conditions would identify which evidence path contributes. Time-weighted and event-weighted trajectory summaries should be compared directly.

Before a larger crossover, preregister the smallest relevant mood-fit effect, sample-size justification, exclusions, primary paired analysis, order analysis, and signal-quality rules [16]. A subsequent dynamic-catalog study should record candidate availability, playback failures, coverage, catalog revisions, explanations, override behavior, and privacy expectations. That sequence separates measurement failure from recommendation failure and tests whether session-aware re-ranking adds value above a conventional preference model.

6 Conclusion & Future Work

Vibe Shuffle studies whether within-listener camera and cardiac changes can inform the next song without being presented as an emotion diagnosis. Personal references center the observations, a listening trajectory preserves their recent movement, and an explicit region-and-distance rule converts the trajectory into a reviewable catalog decision. This answers RQ1 at the level of system design.

The exploratory crossover does not establish improved mood fit or liking. Mood-fit estimates favored Vibe Shuffle, but both outcome intervals included zero, and the incomplete study archive prevents deeper explanation of the differences. The result defines an open empirical question rather than a performance claim.

If later evidence supports the policy, session-aware re-ranking could be examined for exercise, focus, relaxation, games, or accessibility settings in which listening context changes faster than durable preference. Real-world use would require visible uncertainty, an immediate opt-out, local or data-minimizing processing, and adaptation subordinate to the listener's explicit choices.

With validated mappings, complete trial records, preregistered analyses, and a dynamic candidate pool, future work can test the policy as an auditable re-ranking layer above preference-based recommendation. Vibe Shuffle leaves a clear standard for that test: moment-aware music adaptation should be measurable in its effects, inspectable in its decisions, and falsifiable in its claims.

7 Contribution Statement

Markus Baumann, Felix Hajuj, and Miriam Janner jointly contributed to the project concept, study execution, interpretation, and review of the report. Markus Baumann additionally implemented and tested the archived repository version, reproduced the participant-level statistics, and prepared the visual analysis. The available records do not preserve a finer-grained division of writing responsibilities.

Open Science and Transparency

The project repository contains source code, the frozen catalog, tests, figure scripts, participant/session means, and result-provenance records: https://github.com/eybmits/vibe-shuffle. It contains no raw trial rows, sensor streams, recruitment records, administration logs, or code revision tied to the pilot sessions. The computational tests verify the reported source tree, not the historical data-collection environment.

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